CS 401

Sample format for questions for 2nd midterm

Problem 1. . You will be given an instance of the Knapsack problem (the weights and values for each object and the knapsack capacity). You will need to fill out the entries of the dynamic programming table for computing a solution to the Knapsack problem (break ties arbitrarily).

Problem 2. You will be given a statement and asked to state if it is true or false.

Problem 3.

You will be given the statement of a problem with a **very simple** divide-and-conquer solution for it. You will be given a recurrence equation of the running time of the problem.

- You will be asked to justify that the recurrence equation correctly shows the running time of the problem.
- You will be told the final solution of the recurrence equation and then you will be asked to solve the recurrence equation using a specified method (*e.g.*, proof by induction). The solution will not be too complicated and can be worked out by the duration of the exam.

Example: consider the recurrence equation:

$$T(n) = \begin{cases} 5, & \text{if } n = 5 \\ 5 + T(n-5), & \text{otherwise} \end{cases}$$

Assuming that n ***is a multiple of*** 5, *show by induction that* $T(n) = n$.

Solution: if $n = 5$ *then* $T(n) = T(5) = 5 = n$. *Assume by inductive hypothesis that* $T(n) = n$ *for* $n = 5m$. *Consider the next value of* n *that is a multiple of* 5, *namely* $n = 5m + 5$. *Then it follows that* $T(5m + 5) = 5 + T(5m) = 5 + 5m$, *proving our claim.*